

PINAWA, MB, Canada

WMO: 714480

Lat: 50.177N

Lon: 96.065W

Elev: 268

StdP: 98.15

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -33.8	(c) -30.6	(d) -37.5	(e) 0.1	(f) -33.7	(g) -34.1	(h) 0.2	(i) -30.4	(j) 8.4	(k) -13.5	(l) 7.5	(m) -14.1	(n) 1.1	(o) 180	(p) 0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	29.3	20.7	27.6	19.6	26.1	18.6	22.5	27.1	21.1	25.8	20.0	24.4	2.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	16.1	25.3	19.5	14.7	23.8	18.2	13.6	22.5	67.8	26.9	62.8	25.8	58.5	24.5	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.5	6.4	5.6	-38.1	32.8	2.8	1.8	-40.1	34.1	-41.7	35.1	-43.3	36.1	-45.4	37.4
			-38.0	24.7	2.8	1.7	-40.0	25.9	-41.6	26.9	-43.2	27.8	-45.2	29.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-16.6	-14.1	-5.9	3.1	10.3	16.3	19.0	17.9	13.0	5.5	-3.9	-11.8		
	DBStd	13.75	8.60	8.01	8.12	6.08	4.80	3.80	3.09	3.30	4.48	4.82	6.38	7.57		
	HDD10.0	3538	824	674	494	216	57	2	0	0	21	157	416	677		
	HDD18.3	5780	1083	908	752	456	252	82	29	48	169	399	666	936		
	CDD10.0	917	0	0	1	9	65	191	278	246	110	16	1	0		
	CDD18.3	116	0	0	0	0	2	21	49	35	8	1	0	0		
	CDH23.3	1168	0	0	0	3	55	232	452	346	77	4	0	0		
	CDH26.7	254	0	0	0	0	9	51	102	81	12	0	0	0		
(14) Wind	WSAvg	2.5	2.4	2.4	2.6	2.7	2.7	2.3	2.2	2.2	2.4	2.7	2.7	2.4		
(15) Precipitation	PrecAvg	548	22	16	26	32	62	87	79	72	57	44	30	28		
	PrecMax	785	67	45	84	76	133	196	173	162	114	105	90	75		
	PrecMin	344	3	0	3	10	15	19	9	13	10	7	8	7		
	PrecStd	108	13	11	17	16	32	46	36	36	29	27	17	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.5	5.2	15.1	22.9	28.0	31.0	31.7	31.2	28.3	23.2	14.0	4.4		
		MCWB	1.3	2.6	9.7	11.4	16.4	22.5	22.3	21.7	19.1	14.7	8.5	1.8		
	2%	DB	0.9	2.6	10.0	18.9	24.7	27.8	29.2	28.9	25.5	18.8	9.5	1.6		
		MCWB	-0.3	1.1	6.1	9.7	14.0	19.2	21.2	20.3	18.0	12.7	5.6	0.1		
	5%	DB	-1.5	0.7	7.0	15.9	22.1	25.9	27.5	27.0	23.3	15.8	6.6	-0.1		
		MCWB	-2.5	-0.5	3.4	8.3	12.9	17.8	20.0	19.7	16.7	10.9	3.8	-1.0		
	10%	DB	-4.1	-2.0	4.5	13.2	19.4	24.1	25.9	25.2	20.8	13.1	4.1	-1.6		
		MCWB	-4.9	-3.1	1.9	6.9	12.1	17.2	19.2	18.7	15.5	9.5	2.1	-2.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	3.3	10.2	12.8	18.2	24.4	24.2	23.9	21.6	15.6	9.1	2.1		
		MCDB	3.0	5.0	12.9	21.1	23.9	29.4	29.3	28.9	26.1	20.8	13.5	4.1		
	2%	WB	-0.1	1.3	6.0	10.5	16.1	21.1	22.7	21.9	19.2	13.7	6.1	0.4		
		MCDB	0.9	2.6	9.3	17.4	20.9	25.9	27.2	26.7	22.8	18.0	9.1	1.5		
	5%	WB	-2.4	-0.5	3.7	9.0	14.5	19.4	21.4	20.7	17.8	11.6	4.0	-0.9		
		MCDB	-1.4	0.7	6.7	15.2	19.4	23.8	25.9	25.4	21.9	15.1	6.6	-0.1		
	10%	WB	-4.9	-3.1	2.0	7.3	13.2	17.9	20.1	19.5	16.3	9.6	2.2	-2.3		
		MCDB	-4.1	-2.1	4.3	12.8	18.2	22.0	24.5	23.8	19.9	12.8	4.1	-1.5		
(35) Mean Daily Temperature Range	MDBR		10.3	11.9	11.7	11.9	12.1	11.2	11.3	11.6	10.9	8.6	7.3	8.5		
	5% DB	MCDBR	9.8	10.7	12.3	16.6	17.1	14.4	13.4	14.4	14.7	13.8	10.9	9.5		
		MCWBR	8.8	9.0	8.4	9.3	8.6	7.2	6.8	7.3	8.1	8.3	7.4	8.3		
	5% WB	MCDBR	9.5	9.5	11.4	14.7	13.1	11.6	11.8	12.4	11.9	12.4	10.1	9.1		
		MCWBR	8.6	8.1	8.0	8.8	7.7	7.0	6.9	7.3	7.5	8.0	7.2	8.1		
(40) Clear-Sky Solar Irradiance	taub		0.259	0.285	0.294	0.341	0.355	0.377	0.384	0.371	0.342	0.301	0.275	0.249		
	taud		2.274	2.201	2.341	2.388	2.416	2.381	2.381	2.430	2.498	2.576	2.484	2.339		
	Ebn at Noon		787	849	917	904	902	881	868	863	857	843	779	752		
	Edh at Noon		68	95	100	108	110	115	113	103	87	67	56	55		
(44) All-Sky Solar Radiation	RadAvg		1.25	2.42	3.68	4.98	5.40	5.69	5.99	4.95	3.56	2.07	1.12	0.90		
	RadStd		0.13	0.18	0.28	0.38	0.43	0.46	0.30	0.36	0.31	0.27	0.15	0.10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (25 neighbors)	N/A	N/A	N/A	N/A	-1.11	-1.34	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	-0.71	N/A	N/A	N/A	N/A

Nomenclature: See separate page